needs. As we already saw, the master node is the critical node in the cluster and keeps the cluster ticking.

If data is indexed or searched at an exponential growth rate, every node, including the master nodes, takes a performance hit. A slow-performing master node is asking for trouble: cluster operations run slower or can even stall. For this reason, it is always advisable to create a dedicated machine to host the master node. Having a dedicated master node lets clusters run smoothly and mitigates data loss and application downtime.

As mentioned, the rule of thumb is to have at least three dedicated master-eligible nodes in a cluster. When forming a cluster, be sure to set `node.roles` to `master`, as shown in the following snippet, to designate the node-dedicated master:

```
node.roles: [ master ]
```

This way, the dedicated `master` role is not overloaded by performing data- or ingest-related operations and just manages the cluster full time.

That's a wrap! We looked at the administrative part of Elasticsearch in this chapter. In the next chapter, we look at performance tuning.

Summary

- Elasticsearch horizontally scales when adding new nodes to the cluster. The new nodes join the cluster as long as they are associated with the same cluster name and network settings.
- One way to improve read throughput and, thus, performance is to add replicas to the cluster. The replicas take the read hits and serve data quickly.
- Nodes communicate with each other on a transport port, set to 9300 by default. This can be modified by adjusting the `transport.port` property in the elasticsearch.yml file.
- HTTP clients communicate with Elasticsearch on an http.port (set to 9200 by default) using RESTFul endpoints.
- A node can consist of multiple indexes, and each index can consist of multiple shards. The ideal size of a shard should be not more than 50 GB of memory.
- Shards and replicas occupy space, so our organizational strategy should define the appropriate sizes based on current requirements and future use.
- Although adding replicas improves the client's read/query performance, it comes with a cost. We must be sure to carry out appropriate trials and watch for spikes before assigning a standard size for each shard.
- Elasticsearch allows backing up and restoring data using its snapshot and restore feature. Snapshots let us make backups of the cluster to a repository.
- A repository can be a local filesystem or a cloud-based object store such as AWS S3.
- A snapshot can consist of indexes, data streams (as well as the cluster state, such as persistent or transient), indexing templates, index lifecycle management (ILM) policies, and more.